Session – 6

Task-1.

Write down the null hypothesis (H0) and alternative hypothesis (H1) for the following scenario: You believe that people spend more time on Instagram than on YouTube daily.

Solution:

Scenario: You believe people spend more time on Instagram than YouTube daily.

- **H_0 (Null Hypothesis):** People spend the same amount of time on Instagram and YouTube daily (or Instagram ≤ YouTube).
- **H_1 (Alternative Hypothesis):** People spend more time on Instagram than YouTube daily.

Because the claim is specifically **"Instagram > YouTube,"** this is a **one-tailed test**.

Task-2.

Given the data: Group A (users shown a new Zomato home page) had an average order value of ₹350 from 30 users, and Group B (old home page) had an average order value of ₹320 from 30 users. Assume both groups have a standard deviation of ₹50. Calculate the p-value using a two-sample t-test (you may use Python, R, or an online calculator), and decide whether to accept or reject the null hypothesis at a 0.05 significance level.

Hint: Use scipy.stats.ttest_ind in Python or any online t-test calculator if you are not comfortable with code.

Solution:

```
def hypothesis_result(p_value, alpha):
  # Decide on Hypothesis testing
  if p_value < alpha:
    print("Reject the null hypothesis")
  else:
    print("Fail to reject the null hypothesis")


from scipy import stats


# Given Data
mean_A = 350
mean_B = 320
std_A = 50
std_B = 50
```

```
sample_size_A = 30
sample_size_B = 30
alpha = 0.05

# Null Hypothesis: mu_A = mu_B
# Alternate Hypothesis: mu_A != mu_B
# Two-Sample T-Test

# Calculate t-statistic and p-value
# Since we only have summary statistics, use ttest_ind_from_stats()
t_statistic, p_value = stats.ttest_ind_from_stats(
    mean1=mean_A,
    std1=std_A,
    nobs1=sample_size_A,
    mean2=mean_B,
    std2=std_B,
    nobs2=sample_size_B,
    equal_var=True
)

# Print the results
print("T-statistic:", t_statistic)
print("P-value:", p_value)

# Decide on Hypothesis testing
hypothesis_result(p_value, alpha)
```

Output:

```
T-statistic: 2.32379000772445
P-value: 0.023662215835435832
Reject the null hypothesis
```

Task-3.

Pick any feature from your favorite app (like the new dark mode in WhatsApp or a redesigned payment button in Paytm) and describe how you would set up an A/B test for it. Clearly state what your control and variant groups would be, what metric you would measure, and what your null and alternative hypotheses are.

Solution:

Suppose WhatsApp wants to test a **new dark-mode design**.

Control Group (A)

Users see the **existing dark mode**.

Variant Group (B)

Users see the **new redesigned dark mode**.

Metric

Measure **average session duration** or **percentage of users who continue using the app after enabling dark mode**.

Hypotheses

H_0: The new dark mode does not change user engagement.

H_1: The new dark mode increases user engagement.

Test process

1. Randomly divide users into Group A and Group B.
2. Show the existing design to A and new design to B.
3. Collect engagement data for a fixed period.
4. Compare the results statistically.
5. Use the p-value to determine whether the observed difference is statistically significant.

Task-4.

Suppose you run an A/B test for a new 'Add to Wishlist' button design on Flipkart. Out of 1000 users, 520 clicked the new button versus 480 who clicked the old button. Without doing detailed calculations, explain if this difference is likely to be significant or just due to chance. Briefly explain how the p-value helps you decide.

Solution:

Given:

- **New button:** 520 clicks

- **Old button:** 480 clicks
- Total users = 1,000

The new button has a **4 percentage-point higher click rate**:

\[52\% \text{ vs. } 48\% \]

However, the difference **cannot automatically be considered significant** just because 520 > 480.

The **p-value** tells us how likely it would be to observe a difference this large (or larger) if there were actually **no real difference** between the two buttons.

- If **p < 0.05** → statistically significant → reject H_0.
- If **p ≥ 0.05** → not statistically significant → fail to reject H_0.

For this particular example, the difference is relatively small, so **it may reasonably be due to random chance**; a statistical test is needed before concluding that the new button genuinely performs better.